

# CUSTOMER DATA ANALYSIS

CODSOFT INTERNSHIP — TASK 3

Purchasing Behavior • Customer  
Segmentation • High-Value Groups •  
Marketing Insights

## 01 | EXECUTIVE SUMMARY

This report analyzes a 1,000-customer dataset to understand purchasing behavior and identify customer segments that can support targeted marketing decisions. Customers are examined across age, location, customer type, purchase frequency, and product category.

|                                 |                               |                                                                                                                |                                                                                                                      |
|---------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| TOTAL CUSTOMERS<br><b>1,000</b> | TOTAL ORDERS<br><b>10,355</b> | TOTAL SPEND<br> <b>43.08M</b> | AVG. ORDER VALUE<br> <b>4,222</b> |
|---------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|

### Key Findings

- **Hyderabad** is the highest-spending location in the dataset.
- **Fashion** is the highest-spending product category.
- **36-45** is the highest-spending age group.
- **Returning** customers generate the highest total spend among customer types.
- **Very High** customers contribute the highest total spend among purchase-frequency groups.

## 02 | CUSTOMER SEGMENTATION

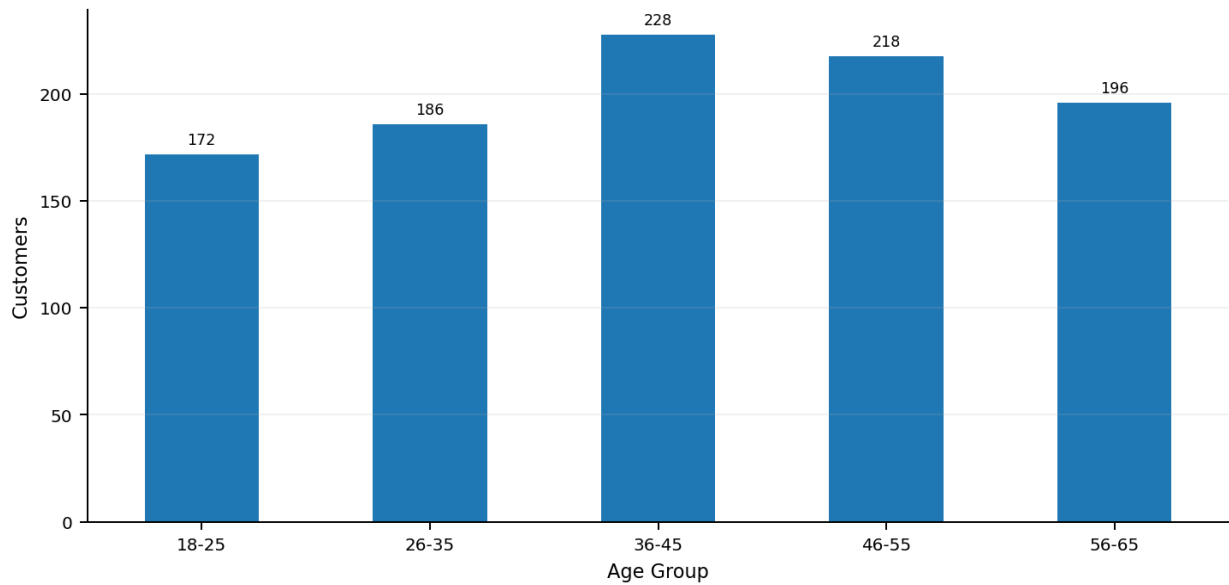
Segmentation helps convert raw customer records into groups that can be compared for customer volume and spending contribution.

| SEGMENT            | HIGHEST-SPENDING GROUP |
|--------------------|------------------------|
| Location           | <b>Hyderabad</b>       |
| Product Category   | <b>Fashion</b>         |
| Age Group          | <b>36-45</b>           |
| Customer Type      | <b>Returning</b>       |
| Purchase Frequency | <b>Very High</b>       |

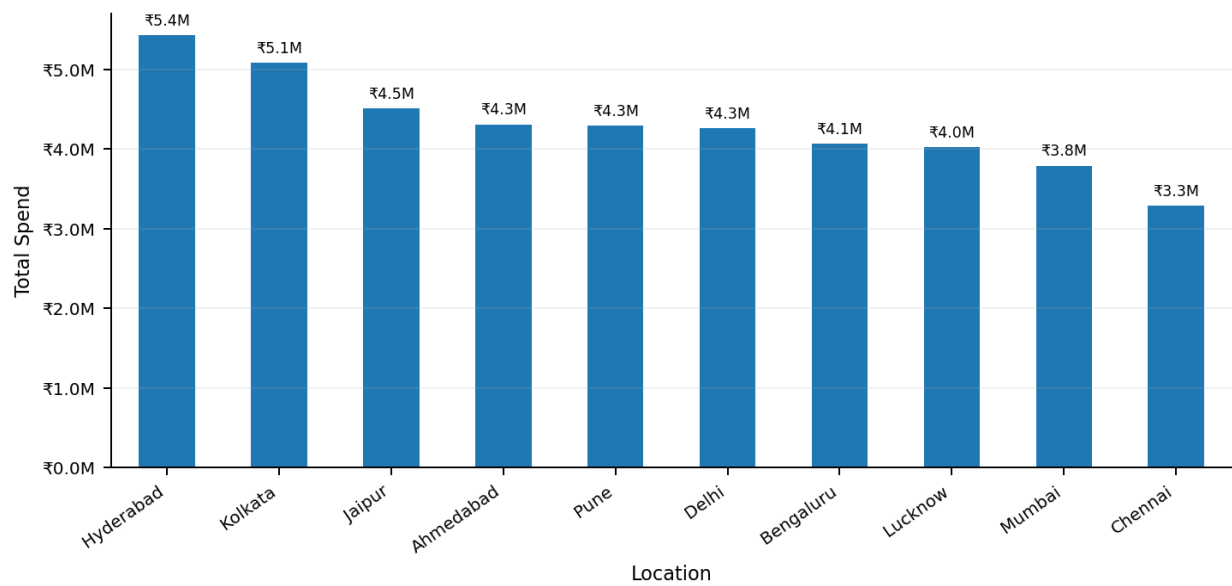
## 03 | VISUAL CUSTOMER INSIGHTS

Age and location provide two important views of customer distribution and value.

### Customer Distribution by Age Group

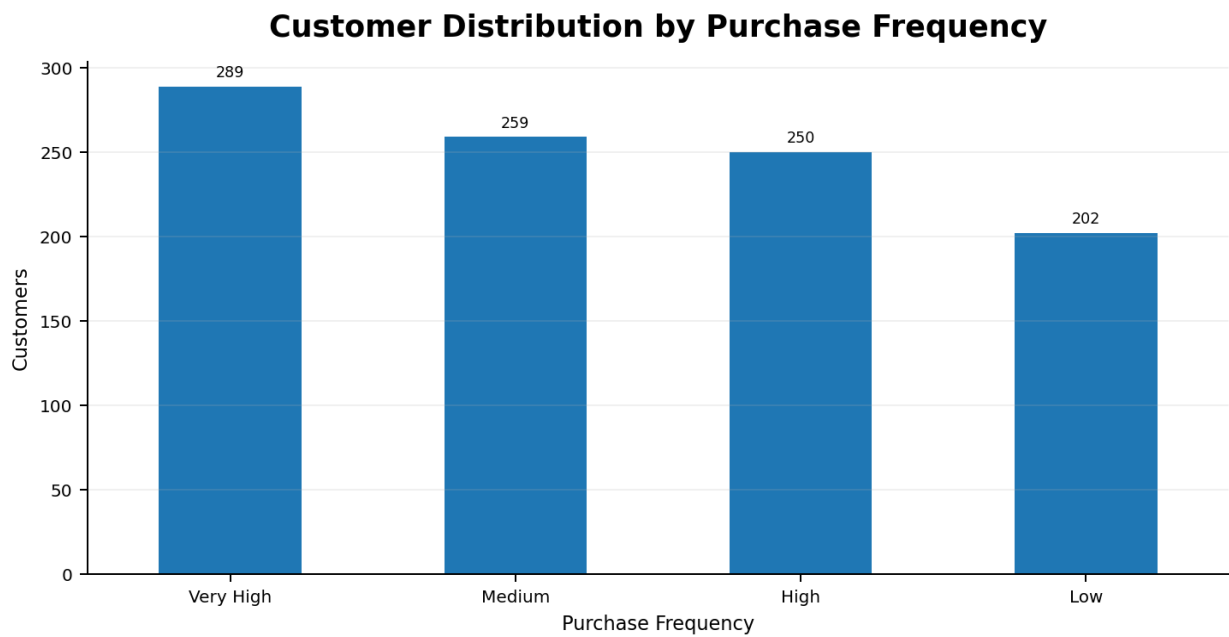
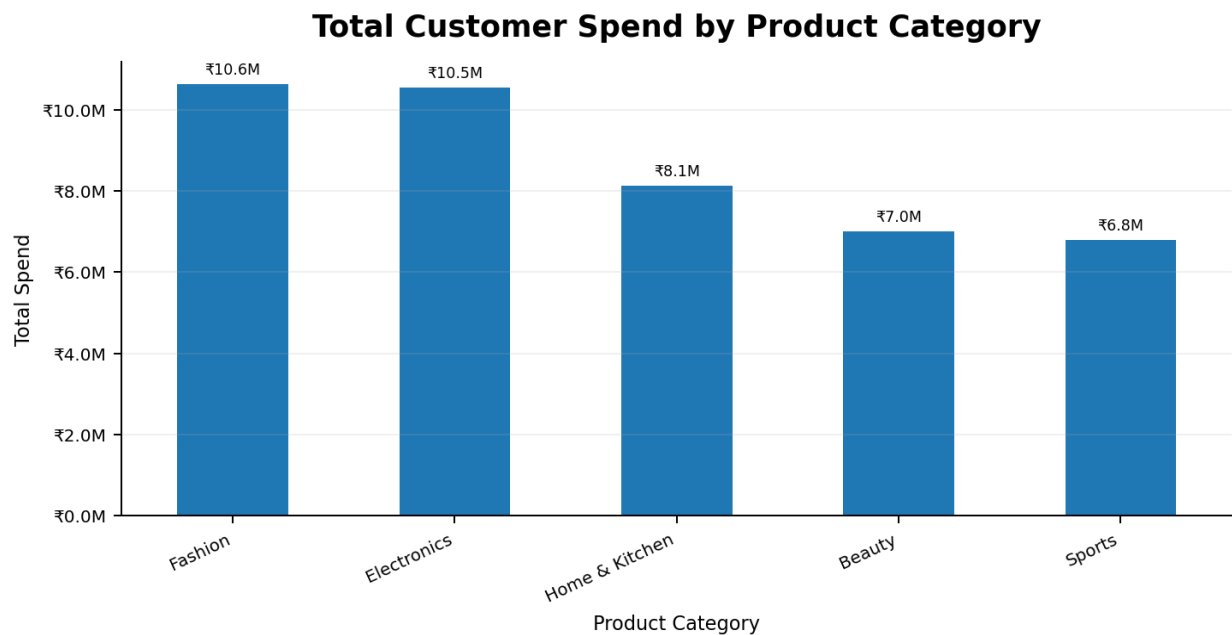


### Total Customer Spend by Location

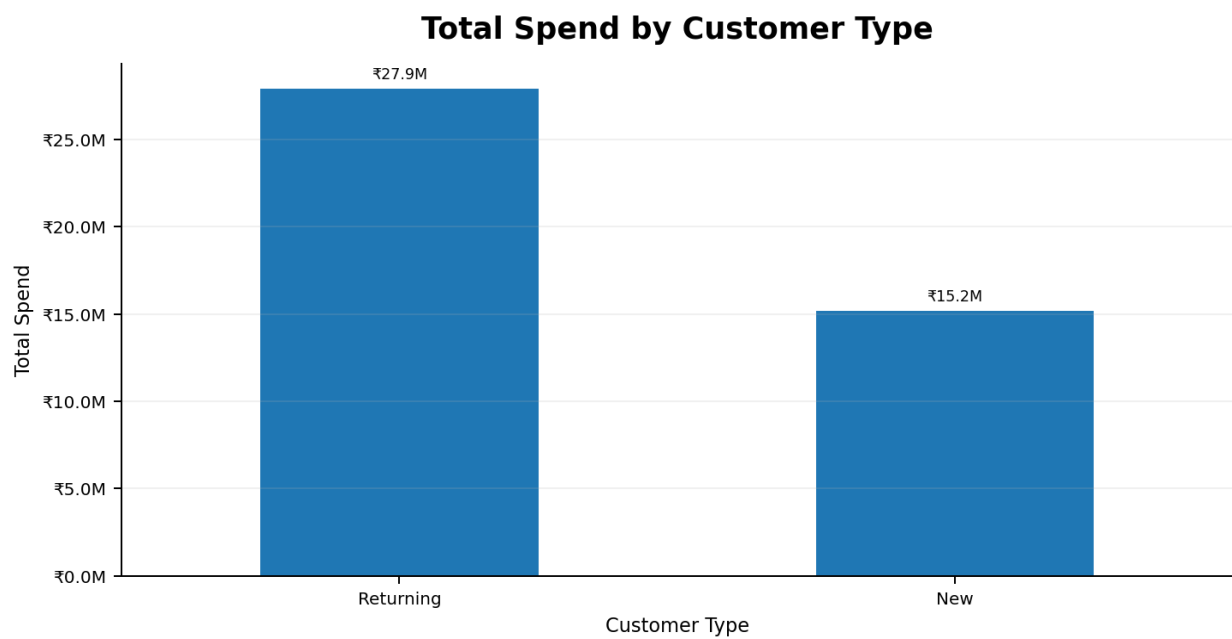


## 04 | PRODUCT & BUYING BEHAVIOR

Product category and purchase frequency highlight where customer value is concentrated.



# 05 | CUSTOMER TYPE ANALYSIS



## Spending by Customer Type

| CUSTOMER TYPE | TOTAL SPEND    | CUSTOMERS |
|---------------|----------------|-----------|
| Returning     | ■27,918,472.86 | 630       |
| New           | ■15,162,118.51 | 370       |

## 06 | BUSINESS INSIGHTS

### Regional opportunity

Hyderabad has the highest total spend, suggesting that regional customer behavior should be examined for campaign planning.

### Category opportunity

Fashion contributes the highest total spend and can be considered for cross-selling, bundles, and personalized recommendations.

### Age opportunity

36-45 is the highest-spending age segment and can be studied for product preferences and tailored messaging.

### Retention opportunity

Returning customers contribute the highest spend among customer types, making retention and loyalty programs relevant.

### Frequency opportunity

Very High customers generate the highest spend among frequency groups; loyalty incentives can help maintain engagement.

## 07 | MARKETING STRATEGIES

- **High-value customers:** introduce loyalty rewards, exclusive offers, and early access to new products.
- **Returning customers:** use personalized recommendations and retention campaigns to encourage repeat purchases.
- **Low-frequency customers:** use targeted promotions, reminders, and limited-time offers to increase purchase frequency.
- **High-spending locations:** focus regional campaigns on locations with strong spending contribution.
- **Top product categories:** use bundles, cross-selling, and personalized recommendations to increase basket value.
- **Age-based campaigns:** tailor promotional messages and product recommendations to different age segments.

## 08 | CONCLUSION

The analysis provides a structured view of customer purchasing behavior and spending patterns. Segmenting customers by age, location, customer type, purchase frequency, and product category creates a practical foundation for targeted campaigns, retention initiatives, and personalized recommendations.

Prepared for portfolio / internship demonstration • Dataset: customer\_data.csv • Analysis: Python & Pandas